Experience:

- Go Outside – Lead Engineer — January 2023 ~ present (contract)
  social networking
  - Supervising, instituting, and enforcing best practices for code.
  - Found, campaigned for, developed, administered, and persuaded management to adopt my recommendations for cloud provider, and backend components.
  - Responsible for our **python** backend, writing all tests – using **pytest** – and keeping them passing.
  - Responsible for all devops on the cloud server, including routine upgrades, SSL certificate, security patches, and regularly negotiating upgrades.
- Jackson Street Capital LLC – Director of Financial Platform Analytics — 2014 ~ 2021 (contract)
  Venture capital
  - Deployed an enhanced dpaste and put it here for investees' use.
  - Inherited an internal tool updating it to use **Spring Boot** – Java – which was enhanced over the years, raising **test coverage** for the various endpoints and keeping them available publicly for users, reducing development time for portfolio companies immeasurably.
  - Social network, a **Ruby on Rails** project, for portfolio companies to communicate/collaborate in a secure, private manner. My responsibilities included its upkeep and automating as much maintenance as possible.
  - Jobs site for startups compensating in equity-only. For some of our startups that wished to have such a list.
  - Initiated, wrote (using **Ruby On Rails**' best practices), deployed on **Heroku**, and maintain a project management tool.
  - Set up a video chat solution, written in **Java** to conduct our conference calls, allowing us to not have to rely on others' infrastructure for often-private communication, using NodeJS to serve **iOS**, **Android**, **Windows mobile** and web clients.
  - Smart address book that automatically recognizes web pages, email addresses, street addresses, and dates in **Ruby on Rails**.
  - Wrote backend and ngrok/tor helper to let one get their bank balances and verify whether one can afford the purchase or not, eventually, planning to add a mobile app and SMS support.
  - Continuous Integration System written in **python**, parsing the junit-formatted XML results file and alerting me to when tests fail to pass by email and SMS (using **Twilio)**.
  - My world clock, so I don't call people at odd hours, with time sourced from PostgreSQL written using **Perl**.
  - CurrencyStack is a **Python** wrapper for CurrencyStack's REST API.
  - HTTP network up-time tool
  - Alexa Skill to read a Twitter feed for a number of seconds, freeing one's eyes to focus on other items of relevance.
  - Using **Python**, LaTeX, and the **Reddit** API, enabled one to publish given any subreddit as a PDF newsletter.
- Zppixbot (acquired by Miraheze) – Backend Engineer March, 2022 ~ June, 2022 (contract)
  Leading provider of chat bots for mediawiki-powered sites like wikipedia
  - Add functionality for currency and unit conversion using **PHP** and **Python**.
- Copilot Travel April 2020 ~ October 2020 (contract)
  Travel-booking site
  - Increase test coverage and maintain a continuous integration/continuous deployment system.
  - Authored a white paper on how multimodal travel could be used to ensure punctuality.
- Fiskkit October, 2015 ~ May 2017 (contract)
  Separating fact from fiction in America's English-language news media
  - Wrote and debugged **Docker** image, reducing new developer on boarding time from days to hours.
  - Maintain NLP stack using SpaCy, and MySQL in **Python**.
  - Twitter notification bot, increasing user engagement with articles by 25%.
  - rewrite our article processing stack using Textblob and Flask to return JSON, persisting using MySQL, hosted on AWS, increasing test-coverage to 100%.
  - Neural network to classify sentences into fact or opinion using RJDBC, Caret, and crowd-sourcing using **R**.
- Fairy Nanny Niñeras – Project Manager — October 2019 ~ December, 2019 (contract)
  Labour play in Yucatan, Mexico
  - Lead website redevelopment effort
- Home is Good LLC September 2018 (contract)
  Proptech play in Austin, Texas
  - **Python** Flask backend for scheduling tours of real-estate.

  - Identifying and diagnosing issues in the **React Native** front end for the above.
  - User invites system in **Python**.
- Trac May, 2018 ~ November, 2018 (contract)
  Accelerate the growth and maximize the success of every musician
  - Evaluate HashGraph API
  - Write backend code in **Go**
- Credfin March, 2017 ~ April, 2017 (contract)
  Leading Australian credit-score company
  - Sped up report generation 99% by moving it from **Django** to a **PostgreSQL** stored procedure allowing for real-time, instead of batch, credit score generation.
- RedRoute LLC, May 2015 ~ November 2015 (contract)
  Ride-sharing start up, acquired by Uber
  - Ported ride sharing startup's backend from **Golang** to **Python** with additional test coverage and reliability enhancements.
- AssistMyCase July 2013 Engineering lead ~ December 2013 (contract)
  Automating paralegal work
  - Supervised named entity classification and co-reference calculation using **Stanford NER** and **GATE**, saving our law firm clients over $200,000 in annual employee expenditure.
  - Managed and mentored junior developers in **Java**.
- Cydesign Labs April 2012 ~ July 2012 (contract)
  ARPA-backed physics-based modeling system to quickly assess design options against requirements during the conceptual design stage
  - Implemented git on Allura using **Turbogears** in **Python**.
  - Devised, designed, specced, and scripted an on-boarding virtual machine image for quality assurance team leading to a 45% decrease in on-boarding time.
- Game Golf (née ActiveMind Technologies) — Data Scientist — San Francisco, CA and Galway, Ireland — August 2011 ~ March 2012 (contract)
  Norwest Ventures-baked makers of a wearable GPS and motion sensing Golf shot tracking System
  - Successfully isolated patterns in geospatial data from golf games using **R**, saving the company over €60,000 in annual salaries.
- Economic Risk Management - Quantitative Analyst — Berkeley, CA — April, 2009 ~ August, 2011 (contract)
  Boutique hedge fund, spun out from Morgan Stanley
  - Automated data downloading using **Perl**.
  - Wrote **Erlang** message-passing system to trigger events – sell positions, manually updating data, generating reports, etc.
  - Designed, installed, and maintained persistence layer architecture in **Java**.
  - Migrated production system from **Google App Engine** to **AWS**, reducing downtime by 30%.
- Quantum 4D - Software Engineer — November 2007 ~ February 2008 (contract)
  Venture-backed service modeling the world economy for the American intelligence community
  - Clean up **Java** code, reducing code size 36%.
  - add test coverage for interaction with **Apache JackRabbit**
- Intellectual Property Litigation Center, Stanford University — Stanford, CA — August 2007 ~ November 2007 (contract)
  - Devised the beginnings of the AI engine that became Lex Machina (acquired by Lexis Nexis) in **Java**.
- Willigent, Needham, MA – March 2002 ~ June 2007 (contract)
  Homeland security contractor
  - Developed and maintained issue tracker using **Java**

Education:

- **Rensselaer Polytechnic Institute** — Troy, NY
  B.S. Computer Science, minors in psychology and economics
  - My undergraduate independent study project was accepted into NetBSD, and later, Apple, as their Bluetooth driver